

Chapter 8

Designing Ambulance Service Districts Under Uncertainty



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Abstract For ambulances, quick response to a medical emergency is critical. Limiting response area for each ambulance may lead to shorter response times to emergency scenes and more evenly distributed workload for ambulances. We propose a two-stage stochastic mixed-integer programming model to address the service district design problem under uncertainty. The proposed model recommends how to locate ambulances to the waiting sites in the service area, and how to assign a set of demand zones to each ambulance at different backup levels. Our proposed Stochastic Service District Design (SSDD) model enables quick response times by jointly addressing the location and dispatching policies in a stochastic and dynamic environment. Each backup level is associated with a given response time threshold. The objective function is to maximize the expected number of covered calls while restricting the workload of each ambulance. The proposed model can be optimized offline as is commonly done for “patrol-beats” used in policing models. We evaluate the implementation of the proposed model via a discrete-event simulation, and compare the model with two baseline policies. Our computational results show a significant improvement in mean response time, reduction of 2 min, and statistically lower average workload of ambulances, of 4% on average, when the proposed model is fully implemented.

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8.1 Introduction

Emergency medical service (EMS) refers to the provision of critical medical care outside of hospitals and transferring of patients to hospitals by equipped vehicles under medical supervision [27]. Each EMS agency operates in a specific service area to respond to emergency calls from the population within its geographic boundaries (i.e., town, city, or county). To better manage demand, the service area may be divided into smaller demand zones [15, 53]. Depending on the regulations and policies of the agency, demand zones may include a number of potential locations, i.e., waiting sites, for the vehicles, medical equipment, and paramedics to be stationed while at rest.

Emergency medical calls are routed through a dispatch center. Dispatchers answering calls interface with a computer-aided dispatch (CAD) system to collect information (e.g., call type, time of dispatch, time of arrival, time the vehicle left the scene and arrived at the hospital, and time the vehicle left the hospital and reported back to service). The service time captures the duration between vehicle dispatch to a call and vehicle return to service. The response time is the difference between time the unit is dispatched and its arrival to the scene.

Quick response to a medical emergency by an ambulance is critical. For example, the survival probability of cardiac arrest patients increases by 50% when attended to by an ambulance within 8 min versus 15 min [51], and it is estimated that survival rate decreases by 5.5% per min that the patient is without treatment [31]. Thus, reducing emergency response times is one of the main ways that EMS systems can decrease mortality of out-of-hospital patients.

In many communities, demand and traffic congestion continue to grow but resources cannot keep pace [25]. Under tight resource constraints, improving operational efficiency is the only way that most EMS systems will be able to maintain or improve quality of service and potentially save more lives. This is a rather difficult task; however, due to several complexities of EMS systems, they are geographically distributed, operate under uncertainty (e.g., calls arrive randomly and service times are stochastic) with heterogeneous customers (e.g., patients with different medical needs), and have inherent interdependencies (e.g., dispatching an ambulance for service impacts the service of future calls).

Due to such uncertain and complex system dynamics, EMS managers face many challenges at the strategic and tactical levels. At the strategic level, they need to decide what locations are potentially appropriate for ambulance waiting sites. This problem has been studied in the literature as a facility location problem [13, 14, 19, 40, 48]. Moreover, it has been shown that limiting or adapting the response area may decrease the average EMS system response time [43] and improve personnel efficiency and satisfaction. Thus, at the strategic level, EMS management may need to investigate the advantages of dividing the original service area into several predefined response districts so as to improve performance of the EMS system. This problem has been handled in the literature as a districting problem [21, 32, 43].

At the tactical level, there are challenges regarding dispatching policies and redeployment of ambulances. A dispatching policy is a set of rules for selecting proper ambulances to respond to emergency calls. This problem has been extensively covered in the literature [33, 44–46]. Ambulance redeployment refers to a strategy that potentially helps to improve EMS performance by repositioning idle ambulances to compensate for unavailability of busy ambulances. Also, redeployment may refer to finding the next location of busy ambulances after service completion (instead of sending ambulances back to their home stations). Many approaches have been proposed in the literature for the redeployment of ambulances [41, 42, 58–60]. Some studies also handle integrated problems in one approach, such as joint location and dispatching problems [23, 66] or joint redeployment and dispatching problems [18].

This chapter concentrates on the development and the evaluation of a two-stage stochastic programming model to address the service district design problem. The proposed Stochastic Service District Design (SSDD) model enables quick response times by jointly addressing the location and dispatching policies in a stochastic and dynamic environment. That is, it first locates ambulances at the potential waiting sites in the service area. Each ambulance is then assigned to a set of demand zones to respond to their emergency calls at different backup levels. Each backup level corresponds to a response time threshold. The proposed SSDD model can be optimized *offline* and the outputs regarding the dispatching policy are easily applied in a dynamic manner. Such an implementation is similar to the idea of “patrol-beats” used in policing models and it reduces the technological capabilities required to integrate the model within basic computer-aided dispatch (CAD) systems.

In the remainder of this chapter, we briefly review the related literature in Sect. 8.2. We present the model formulation in Sect. 8.3. We discuss the output representation and the numerical results of implementing the proposed model via simulation in Sect. 8.4. We finally conclude in Sect. 8.5.

8.2 Literature Review

EMS providers seek to minimize response times and maximize coverage, i.e., the proportion of calls answered within a given time threshold. The EMS literature has addressed four different problems with the goal of improving service: ambulance location/allocation, redeployment, and dispatching problems as well as the districting problem. These problems are formulated and solved in three (not necessarily mutually exclusive) streams: static/deterministic, dynamic/real-time, and probabilistic. Earlier approaches mostly dealt with static and deterministic models for location/allocation problems [13, 65]. Static models are proposed for the strategic decisions in the planning stage [10]. Dynamic and real-time approaches address tactical decision such as ambulance redeployment [27]. Probabilistic approaches handle the uncertainty of ambulance availability. Brotcorne et al. [10], Goldberg [22], Li et al. [37], Başar et al. [5], Bélanger et al. [7], and Aringhieri et al. [4] reviewed mathematical models in the EMS literature. The remainder of this section

reviews studies addressing ambulance location/allocation as well as ambulance dispatching and redeployment, with a focus on those works that address uncertainty. We also summarize the studies on service districting problem in the EMS literature as well as in the literature of other applications.

8.2.1 Ambulance Location/Allocation

While there is a large body of research related to resource location/allocation in EMS systems, there is a lack of mathematical models that explicitly consider uncertainty. The maximum expected covering location problem (MEXCLP) was formulated as one of the first probabilistic models for the ambulance location problem [14]. MEXCLP assumed that each ambulance has the same probability, called the *busy fraction*, of being unavailable to respond to a call, and all ambulances are independent. Employing the busy fraction concept, the maximum availability location problem was proposed to maximize the demand covered with a given probability [54]. Several studies viewed ambulances as servers in a queuing system [6, 8, 11, 39, 40], and applied the hypercube model of Larson [32] to estimate busy fractions associated with the whole system or with a specific waiting site. Furthermore, compliance tables are designed to prescribe ambulance allocation given a certain fleet size. A Markov model was proposed to estimate the busy probabilities of ambulances for a given compliance table [1]. Optimizing a compliance table design was first proposed by Sudtachat et al. [63]. Another way to deal with uncertainty is through detailed simulation models, such as Andersson and Varbrand [2] and Sudtachat et al. [62], but these can only be used to compare proposed policies or heuristics, not to find optimal solutions, and cannot be generalized to systems under different operating conditions.

8.2.2 Ambulance Dispatching and Redeployment

Advances in CAD systems and access to data have spurred research in real-time models that are executed *online*. These models focus on dispatching and redeploying ambulances based on the current system status and demand forecasts [41, 53, 58, 59]. A mixed-integer program was formulated to provide a dynamic relocation strategy that maximizes the expected demand coverage while controlling the number of waiting site relocations [20]. System-status management models consider real-time decisions and not strategic decisions. Real-time recommendations must still adhere to the underlying design and response policies [42]. Advanced CAD systems may be linked with GIS technology to make dispatch recommendations based on real-time traffic information and demand forecasts. However, suburban and rural EMS systems typically have basic CAD systems that lack GIS capabilities. Even with advanced CAD systems, enhancing their capabilities is expensive and some agencies

view this as an avoidable cost [56]. Thus, it is important to provide guidelines for data-driven optimization models that can be integrated into existing CAD systems.

8.2.3 Service District Design

Service district design is a strategic decision that is made *offline* for constructing the primary (and possibly backup) response area(s) of ambulances to improve coverage and enable quick response times. This problem can be addressed from two complementary perspectives. The first perspective involves grouping demand zones into compact and contiguous districts such that the set of assigned vehicles to each district is primarily responsible for responding to calls within that district. This is the traditional districting design approach which creates clear physical borders for districts. In geographically compact districts the distance to the center does not vary widely, and contiguity ensures that it is possible to travel between any two points within a district without having to leave the district. A mixed-integer linear program was formulated to optimize weighted objectives of district balance (defined as relative deviation of the district size from the average district size), and recommend compact and contiguous districts for political districting, police patrol area delineation, and sales territory design applications [30]. See additional examples in the literature of political and public service districting problems [12, 47, 64]. Modeling such a districting design would result in highly symmetric mathematical programs in which several equivalent solutions can be obtained by changing the “center” of a district plan to any of the demand zones included in the district. Therefore, solution algorithms developed for this modeling perspective have to explore many alternative symmetric solutions which increases the computational burden [61]. The problem of dividing an area into a set of compact and contiguous districts is NP-complete [52]. The existing models and solution approaches in the literature are limited to solve only small instances [16, 57], or some common districting criteria (i.e., contiguity, compactness, balanced districts) are not fully enforced in the formulation [36, 50, 55]. Considering uncertainty in the EMS systems is even more challenging in such districting models. The second perspective to district design assigns a set of demand zones to each vehicle at different backup levels. This approach differs from designing contiguous and compact districts as it prescribes how to dispatch vehicles to calls in real time without explicitly partitioning the response area into geographical districts. This chapter adopts the latter perspective to address the districting problem in EMS systems with uncertainty considerations.

There have been few works in the literature related to district design of EMS systems. An optimization framework was developed using genetic algorithms to district a region and deploy ambulances on highways to minimize the average response time; however, the results were limited to a linear service region [26]. A heuristic was proposed to design districts for an EMS system, and developed a simulation model to evaluate the performance of several integrated districting

and dispatching policies in terms of patient survival probability [43]. A mixed-integer program was formulated to locate and dispatch ambulances through district design [3]. They consider the uncertainty in ambulance availability and travel times through a hypercube model approximation.

The districting problem has been extensively studied in various other applications such as geography, politics, ecology, business, and public service. A broad perspective on common criteria and limitations in districting applications was provided [28]. Not many works in the literature have studied stochastic districting problems. A two-stage stochastic program was developed for vehicle routing and districting problem with stochastic customers in which districting decisions are made in the first stage and the routing expenses are approximated in the second stage [34]. They addressed the compactness of districts in the objective function and developed a large neighborhood search heuristic to solve the model. Also, the districting and routing problem were addressed with stochastic demand using a two-stage stochastic program and developed Tabu search and multistart heuristics [24].

The proposed stochastic programming model in this chapter is the first analytic model that explicitly considers uncertainty for the EMS service district design problem. The proposed formulation can model complex EMS system dynamics while leveraging historical data. Only a few papers considered stochastic programming formulations for EMS problems. A two-stage stochastic programming model was developed that is solved repeatedly on a rolling horizon basis in real time to minimize the number of relocation moves during the planning horizon, while maintaining an acceptable service level [49]. Furthermore, a two-stage stochastic programming model with chance constraints was formulated to locate ambulances in the first stage, and allocate demand points to ambulances in the second stage [8]. However, their proposed model does not consider the uncertainty in the availability of ambulances when evaluating coverage. In a similar vein, a chance-constrained optimization to relocate ambulances was proposed [38]. In summary, there is a lack of mathematical models for spatially distributed service systems that explicitly consider uncertainty which is an endemic feature of such systems.

8.3 Model Formulation

Better service district designs may lead to shorter response times, more equitable division of the workload, familiarization with the assigned area, and more efficient use of personnel. The average response time and the variation of workload among different districts are two important performance measures of service district design. Limiting the primary response area of ambulances enables the EMS system to decrease the average response time of paramedic support to the scene [43]. However, a service district design, just like any other decision for improving operational efficiency in EMS systems, has to consider uncertainty as EMS systems deal with uncertain demand with several stochastic attributes such as arrival time, location, and service time of emergency calls [18].

We generate a finite set of scenarios to include the uncertainty of the emergency call sequence into the decision-making process. Each scenario consists of a set of calls that occurred during a shift and call specifications, i.e., location, arrival time, and service time. We assume that the likelihood of an ambulance being unavailable to respond to a call (i.e., busy fraction) is identical and independent for all ambulances under each scenario [14].

We formulate a two-stage stochastic programming model to design EMS service districts with the goal of maximizing expected number of emergency calls that are responded to in a timely manner while limiting the workload of ambulances. Each district consists of multiple demand zones, and the ambulance assigned to a district is responsible to serve calls in those demand zones within a certain response time threshold. In the first stage, ambulances are assigned to waiting sites, and demand zones are assigned to ambulances at different backup levels. Each backup level is associated with a response time threshold. The first-level backup ambulance of a demand zone primarily responds to calls from that zone; however, if the first-level backup ambulance is busy, then the second-level backup ambulance responds, and so on. The second-stage problem measures the expected proportion of calls that are served by each ambulance at all backup levels as a result of the first-stage assignments. The proposed model in fact prescribes how to dispatch ambulances to emergency calls depending on their real-time availability at different backup levels without explicitly partitioning the service area into compact and contiguous districts.

Let $\mathcal{I}$ be the set of demand zones, $\mathcal{J}$ be the set of waiting sites ($\mathcal{J} \subseteq \mathcal{I}$), and $\mathcal{R}$ be the set of backup levels, i.e., $\{1, \dots, R\}$. We define $\mathcal{J}_{ir}$ as the set of waiting sites that can serve demand zone $i \in \mathcal{I}$ at the $r^{th} \in \mathcal{R}$ backup level (i.e., $\mathcal{J}_{ir}$ includes those waiting sites that are located within the given time threshold defined for the r^{th} backup level from demand zone i). The number of available ambulances is denoted by κ . The first-stage decision variables are denoted by:

- $x_j \in \{0, 1\}$: equals 1 if an ambulance is located at site $j \in \mathcal{J}$, and 0 otherwise.
- $y_{ijr} \in \{0, 1\}$: equals 1 if the $r^{th} \in \mathcal{R}$ backup ambulance for demand zone $i \in \mathcal{I}$ is located at site $j \in \mathcal{J}$, and 0 otherwise.

Let $\mathcal{X} := \{x_j, j \in \mathcal{J}\}$ and $\mathcal{Y} := \{y_{ijr}, i \in \mathcal{I}, j \in \mathcal{J}, r \in \mathcal{R}\}$. The Stochastic Service District Design (SSDD) model is then formulated as:

$$(\text{SSDD}) \quad Z^* = \max \mathbb{E} (Q_s(\mathcal{X}, \mathcal{Y})) \quad (8.1a)$$

$$\text{subject to} \quad \sum_{j \in \mathcal{J}_{ir}} y_{ijr} \geq 1 \quad i \in \mathcal{I}, r \in \mathcal{R} \quad (8.1b)$$

$$\sum_{r \in \mathcal{R}} y_{ijr} \leq x_j \quad i \in \mathcal{I}, j \in \mathcal{J} \quad (8.1c)$$

$$\sum_{j \in \mathcal{J}} x_j \leq \kappa \quad (8.1d)$$

$$x_j, y_{ijr} \in \{0, 1\} \quad i \in \mathcal{I}, j \in \mathcal{J}, r \in \mathcal{R} \quad (8.1e)$$

The first-stage objective function (8.1a) maximizes the expected recourse function which will be elaborated in the second-stage problem. Constraints (8.1b) state that demand zone $i \in \mathcal{I}$ must be covered by at least one ambulance located at covering site $j \in \mathcal{J}_{ir}$ at backup level $r \in \mathcal{R}$. Constraints (8.1c) ensure that site $j \in \mathcal{J}$ responds to calls from demand zone $i \in \mathcal{I}$ at any backup level $r \in \mathcal{R}$ only if there is an ambulance located at site $j \in \mathcal{J}$. Constraint (8.1d) enforces the fleet size limit.

Let $\mathcal{S}$ be the set of generated scenarios. The second-stage decision variable is defined as follows:

- $z_{ij}^s \in [0, 1]$: the proportion of calls from demand zone $i \in \mathcal{I}$ that are served by an ambulance at waiting site $j \in \mathcal{J}$ under scenario $s \in \mathcal{S}$.

Table 8.1 summarizes the notation of parameters used in the second-stage problem of the SSDD model. The second-stage problem under scenario $s \in \mathcal{S}$ is given by:

$$Q_s(\mathcal{X}, \mathcal{Y}) = \max \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} d_i^s z_{ij}^s \quad (8.2a)$$

$$\text{subject to } z_{ij}^s \leq \sum_{r \in \mathcal{R}} \rho_{ir}^s y_{ijr} \quad i \in \mathcal{I}, j \in \mathcal{J} \quad (8.2b)$$

$$\sum_{i \in \mathcal{I}} d_i^s z_{ij}^s \leq F^{\max} x_j \quad j \in \mathcal{J} \quad (8.2c)$$

$$z_{ij}^s \geq 0 \quad i \in \mathcal{I}, j \in \mathcal{J} \quad (8.2d)$$

The second-stage objective function (8.2a) maximizes the coverage. Constraints (8.2b) compute the proportion of calls in demand zone $i \in \mathcal{I}$ that are

Table 8.1 Parameters in the second stage of the service district design model

ρ_{ir}^s	Expected proportion of calls from demand zone i under scenario s that arrive when the first $(r - 1)$ backup ambulances are busy and the r^{th} backup ambulance is available: $\sum_{r \in \mathcal{R}} \rho_{ir}^s \leq 1, i \in \mathcal{I}, r \in \mathcal{R}, \text{ and } s \in \mathcal{S}$
d_i^s	Ratio of calls from demand zone $i \in \mathcal{I}$ to all calls during the planning horizon under scenario $s \in \mathcal{S}$
$F^{\max}$	Maximum proportion of all calls that can be responded by each ambulance

responded to from site $j \in \mathcal{J}$ at different backup levels. Note that parameter ρ_{ir}^s is a function of the busy fraction probability of ambulances and the sequence of calls generated from a demand zone under each scenario. Constraints (8.2c) establish the upper bound on the proportion of all calls that can be responded to by site $j \in \mathcal{J}$ at which an ambulance is located. We estimate parameter $F^{\max}$ in constraints (8.2c) by:

$$F^{\max} = \frac{\max_{s \in \mathcal{S}} (\sum_{i \in I} d_i^s)}{\kappa} \quad (8.3)$$

Equation (8.3) sets the upper bound for the workload of each ambulance by equally distributing the calls between ambulances, i.e., we divide the maximum total number of calls during a shift generated in scenario set $\mathcal{S}$ by the number of available ambulances. Finally, constraints (8.2d) enforce the second-stage decision variables to be nonnegative.

8.4 Computational Results

We discuss the model output on two different instance sets:

1. Small instances: configuration of a service area is randomly generated. Emergency calls are uniformly distributed in the service area and their arrival times and service times are also generated randomly from a uniform distribution. All demand zones are considered as potential waiting sites for ambulances ($|I| = |J|$).
2. Large instance: scenarios are generated based on a real dataset collected from an EMS agency in Mecklenburg County, North Carolina. The service area is about 540 square miles that is hypothetically divided into 168 demand zones (i.e., $|I| = 168$). Each zone is 4 square miles (2×2) [17]. The set of waiting sites J is assumed to be the centroid of each demand zone (i.e., $|J| = 168$) because ambulances are allowed to wait at any location in Mecklenburg County. The dataset includes about 50,000 calls in Mecklenburg County for 650 shifts in 2004. The spatial demand distribution is highly concentrated in the central demand zones and is more sparse in the peripheral areas. Furthermore, demand significantly varies based on day of the week and time of the day [17]. We assume that each shift in the data corresponds to a scenario.

In both instances, we consider three different backup levels, $|R| = 3$, with corresponding coverage thresholds of 8, 12, and 16 min. We solve the extensive form of the proposed SSDD model (8.1). All computational experiments have been performed using CPLEX 12.6 on a Windows computer with 32 GB of memory and 3.4 GHz Intel Core i7 processor. Section 8.4.1 presents the model outputs on small examples. Section 8.4.2 evaluates the solution of the large instance via simulation.

8.4.1 Discussion of the Model Output

The SSDD model outputs the optimal waiting sites for available ambulances. Also, it determines the response design at all backup levels corresponding to each ambulance site. We present the model outputs on two different small examples in Figs. 8.1 and 8.2. In each figure, demand zones with asterisks indicate the waiting site at which an ambulance is located. Demand locations shown in dark gray, light gray, and white illustrate the zones for which the ambulance located at the asterisked site serve as the first, second, and third backup, respectively. Demand zones shown in red are not covered within any backup level of the corresponding ambulance at the asterisked site. The service area in both examples is a random subset of zones selected from the service area of the EMS agency at Mecklenburg County.

The first small example is shown in Fig. 8.1 which is the ambulance response design for a service area including 15 demand zones (i.e., $|I| = 15$), 4 ambulances (i.e., $\kappa = 4$), and 3 backup levels (i.e., $|R| = 3$). Figure 8.1a–d shows the response design at each backup level for ambulances stationed at waiting sites 3, 7, 9, and 11, respectively. This result is the optimal solution of the SSDD model for 200 scenarios in which the busy fraction of ambulances ranges between 41.6% and 64.3% during a 6-h shift. In this instance, the computational time to solve the extensive form of the SSDD model by CPLEX is 20 s. Figure 8.1 demonstrates that all demand zones are covered at least once within each backup level.

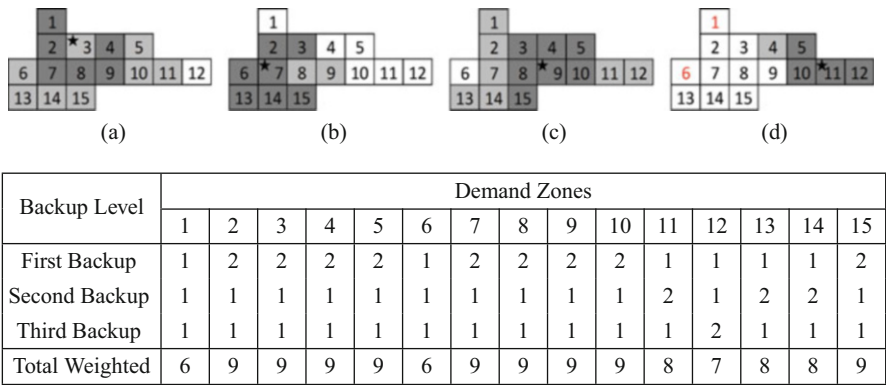
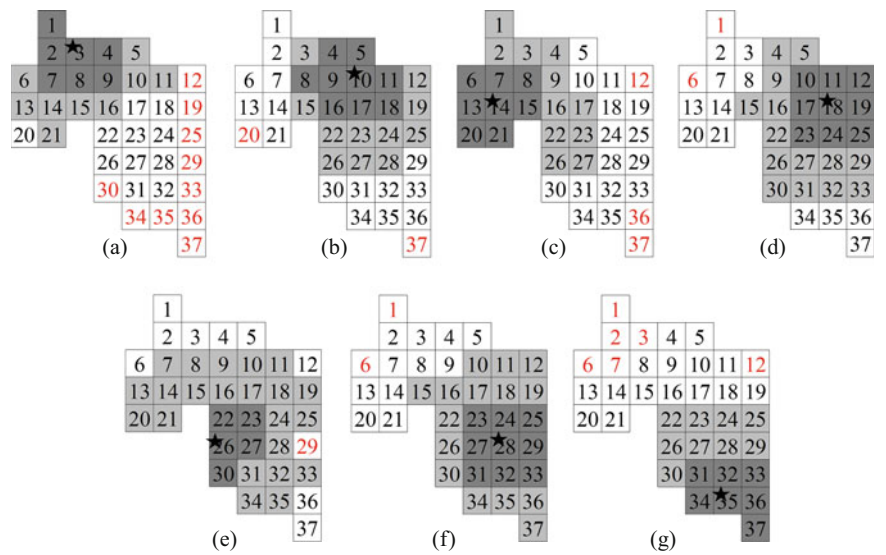


Fig. 8.1 Ambulance response design for a small example with 15 demand zones, 4 ambulances, and 3 backup levels. Demand zones with asterisks indicate the station at which an ambulance is located. Dark gray and light gray demand zones are covered within the first and the second backup levels by the corresponding ambulance located at the asterisked site, respectively. White zones demonstrate the third backup level. Demand zones shown in red are not covered within any backup level of the corresponding ambulance at the asterisked site. The table shows the number of ambulances covering at each backup level as well as the total weighted coverage for each demand zone. The first, second, and third backup levels weigh 3, 2, and 1, respectively



Backup Level	Demand Zones																		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
First Backup	1	1	1	2	1	1	2	3	2	2	2	1	1	1	1	1	2	2	1
Second Backup	1	1	2	2	2	1	1	1	3	3	3	2	2	2	5	5	3	2	3
Third Backup	2	4	3	3	4	2	3	3	2	2	2	1	4	4	1	1	2	3	2
Total Weighted	7	9	10	13	11	7	11	14	14	14	14	8	11	11	14	14	14	13	11

Backup Level	Demand Zones																		
	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	
First Backup	1	1	1	3	2	2	1	2	1	1	1	2	2	2	1	1	1	1	
Second Backup	1	2	5	3	3	3	5	4	4	2	3	2	2	2	2	2	1	1	
Third Backup	4	4	1	1	2	1	1	1	2	2	2	3	3	2	3	3	3	2	
Total Weighted	9	11	14	16	14	13	14	15	13	9	11	13	13	12	10	10	8	7	

Fig. 8.2 Ambulance response design for small example with 37 demand zones, 7 ambulances, and 3 backup levels. Demand zones with asterisk indicate the station at which an ambulance is located. Dark gray and light gray demand zones are covered within the first and the second backup levels by the corresponding ambulance located at the asterisked site, respectively. White zones demonstrate the third backup level. Demand zones shown in red are not covered within any backup level of the corresponding ambulance at the asterisked site. The table shows the number of ambulances covering at each backup level as well as the total weighted coverage for each demand zone. The first, second, and third backup levels weigh 3, 2, and 1, respectively

Some of the demand zones may not be served by an ambulance at a specific site as shown in Fig. 8.1d in red fonts, i.e., demand zones 1 and 6 are not covered by the ambulance located at station 11 at any backup level. However, this does not mean that those demand zones are not covered at all. According to Fig. 8.1, zone 6 is covered by the ambulances located at sites 3, 7, and 9. Furthermore, it is possible that the ambulance located at a site does not cover the corresponding demand zone at the first backup level. For example, in Fig. 8.1a the ambulance located at site 3 covers demand zone 3 at the second backup level. This can happen as a result of the workload constraint and may be difficult to justify in practice. Such circumstances do not occur frequently according to our trial runs, thus we employ a correction to the model output and assume if an ambulance is located at a site, it must always cover the corresponding demand zone at the first backup level.

Figure 8.2 illustrates the solution to the second small instance which includes 37 demand zones (i.e., $|I| = 37$), 7 ambulances (i.e., $\kappa = 7$), and 3 backup levels (i.e., $|R| = 3$). Figure 8.2a–g shows the response design at each backup level for ambulances stationed at sites 3, 10, 14, 18, 26, 28, and 35, respectively. This instance includes 200 scenarios and the busy fraction of ambulances ranges between 51.2% and 75.3% during an 8-h shift. The time required to solve the extensive form of the SSDD model is 72 s.

As demonstrated via two examples in this section, the SSDD model (8.1) is flexible to find the optimal solution to different EMS systems with different demand congestion levels and shift lengths. A drawback of our model is the possibility of an infeasible solution due to the constraints (8.1b) that force the system to cover each demand zone at least once within each backup level. For instance, it was necessary to have at least 4 and 7 ambulances, respectively, in the first and the second small example to avoid infeasibility. We do not recommend to relax the constraints (8.1b), although this can guarantee feasibility. Some demand zones might not be covered even within the first backup level after relaxing these constraints which is not practical.

8.4.2 Model Evaluation

We develop a discrete-event simulation to evaluate the performance of the SSDD solution to the large instance based on the real dataset. We generate emergency calls to the Mecklenburg EMS agency during 2 weeks, i.e., 28 12-h shifts. We consider 18, 19, and 20 ambulances and sample 450 scenarios. This fleet size is the common range of available ambulances in Mecklenburg county. All reported instances are solved optimally within 4 h. We could not increase the scenario size more than 450 due to the computational intractability of solving the extensive form of the model. For instance, when the number of scenarios is 500 and the fleet size is 20, the best reported gap by CPLEX is 27.5% after 4 h run time. A tailored decomposition algorithm might be developed to solve large scale instances rather than solving the extensive form of the model. This is beyond the scope of this

chapter, and we refer the reader to Enayati et al. [18] and Lei et al. [35] for more information on decomposition methods in a similar application domain. We assume that ambulances are initially located at the sites proposed by the SSDD model and they return to their home sites after service completion. Ambulances can be dispatched to new calls before returning to their home sites. Furthermore, if more than one ambulance is available for a call, the closest one is dispatched. We compute actual road network distance for all ambulance trips using the Matlog toolbox in MATLAB [29].

It might not be possible to apply optimal dispatching decisions in real life due to legal regulations that mandate sending the closest available ambulance to an emergency call. Hence, we study the isolated effect of ambulance location and dispatching decisions on the performance of an EMS system by considering two additional baseline policies. The *baseline policy-1* merely uses the ambulance sites in the solution to the SSDD model. In this baseline policy, ambulances are dispatched according to the myopic policy that sends the closest available ambulance to an emergency call. In *baseline policy-2* ambulances are initially located based on the Maximum Covering Location Problem (MCLP) [13]. We use the historical proportional demand corresponding to each demand zone as the weight multiplier in the objective function of the MCLP model. Dispatching policy in the *baseline policy-2* also sends the closest available ambulance. We report two performance measures along with the computational time to solve the extensive form of the SSDD model in Table 8.2:

- MRT (min): mean response time to calls during each shift.
- Workload (%): average percentage of time that an ambulance is busy with service and travels (dispatching and transportation to hospital) during each shift.

Table 8.2 95% confidence intervals of mean response time and workload measures for 2-week simulation of an EMS system

	SRDP	Baseline-1	Baseline-2
	$\kappa = 18$		
MRT (min)	[9.9, 13.7]	[11.3, 15.1]	[11.9, 15.8]
Workload (%)	[54.8, 59.6]	[58.9, 62.6]	[59.5, 63.2]
Time (s)	7982	–	–
	$\kappa = 19$		
MRT (min)	[9.2, 13.3]	[10.9, 14.9]	[11.2, 15.1]
Workload (%)	[50.2, 52.8]	[53.3, 56.5]	[53.6, 56.2]
Time (s)	8235	–	–
	$\kappa = 20$		
MRT (min)	[8.8, 12.6]	[10.5, 14.0]	[10.6, 14.9]
Workload (%)	[43.2, 45.7]	[47.1, 49.3]	[47.8, 49.7]
Time (s)	9998	–	–

The reported time indicates the computational time to obtain the solution of solving the extensive form of the SSDD model with 450 scenarios by CPLEX

We make the following observations from Table 8.2:

- Full implementation of the SSDD model always outperforms both baseline policies in terms of mean response time. Comparing to the *baseline policy-2* in which alternative location and dispatching policies are applied, the full implementation of the SSDD model improves the mean response time by 2 min on average. We also observe that solely implementing the SSDD location solution with an alternative dispatching policy, as in *baseline policy-1*, does not improve the mean response time as much.
- Full implementation of the SSDD model always imposes less workload on ambulances by about 4% on average as compared with both baseline policies. The workload of ambulances in *baseline policy-1* and *baseline policy-2* is statistically similar.

Finally, we calculate the value of stochastic solution that measures the potential benefit of solving the stochastic program over solving a deterministic program in which random parameters are replaced with their expectations [9]. That is, we generate a single scenario in which all random parameters (e.g., d_i^s) are estimated to the average of all 450 generated scenarios. We then run the SSDD for this scenario and evaluate the output using simulation. The value of stochastic solution in our set of experiments is reported 18% and 29% based on the MRT and the workload measures, respectively.

8.5 Conclusion

In this chapter, we formulate a two-stage stochastic program to address the service district design problem. This model presents two outputs: (1) available ambulances are located at the potential sites in the service area, and (2) response districts for each ambulance are designed at different backup levels (i.e., each ambulance is assigned to serve a set of demand zones within a certain response threshold). We maximize the expected proportion of covered calls while restricting the workload of ambulances. The proposed model assumes that each ambulance has identical and independent busy fraction probability.

We illustrate the solution on two small examples. We then evaluate the implementation aspects of the proposed model via a discrete-event simulation by comparing the model with two baseline policies to study the isolated impact of location and dispatching decisions resulting from the proposed model. The two baseline policies are designed due to the mandatory regulation in most EMS systems that obligates to send the closest available ambulance to an emergency call. Our computational results show a significant improvement in mean response time by 2 min on average when implementing the proposed model. We also observe that average workload of ambulances is statistically less by 4% on average when the model is fully implemented.

Future studies may formulate a model that endogenously estimates the busy fraction of ambulances as a decision variable for the service district design problem. The results of such a formulation can be compared with the proposed model in this chapter to study the impact of incorporating the ambulance busy fractions exogenously versus endogenously on the quality of solutions. An endogenous formulation could also alleviate the possibility of infeasible solutions. Furthermore, future studies may develop a decomposition algorithm for the proposed two-stage stochastic program to deal with computational intractability due to large number of scenarios.

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